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CyberSecurity

1 - "Introduction to Computer Security"

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CyberSecurity

1 - "Introduction to Computer Security"

This block of slides does not refer to a specific chapter in the book.

It is a "gentle introduction"

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Outline

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A few basic principles of computer security:

- principle **1**: "*non-existence of secure systems*"
- principle **1 (corollary)**: "*systems complexity*"
- principle **2**: "*entities composing a system*"
- principle **3**: "*security = knowledge*"

Security of a cryptosystem (*i.e.*, *security through obscurity*)

Principle 1: on "Secure Systems"

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SECURE SYSTEMS DO NOT EXIST

- **The software is often far from perfect**. It is full of **implementation bugs** and **design errors**. It is the same for **communication protocols**
- **The "unbreakable system" is only a myth**. Just like the impenetrable bank vault or the unsinkable boat (*e.g., RMS Titanic*)

Principle 1: on “Secure Systems”

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Un attaccante attacca un sistema se c'è abbastanza valore che ne vale la pena l'attacco

SECURE SYSTEMS DO NOT EXIST

Security is costly: so there are various level of security

The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

È molto difficile valutare correttamente il livello di sicurezza necessario per un sistema.

Le vulnerabilità possono essere scoperte (da una persona pagata o criminale)

Principle 1: on "Secure Systems"

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Quanto sono disposto ad investire per trovare/risolvere le vulnerabilità del mio sistema?

SECURE SYSTEMS DO NOT EXIST

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

What are we protecting?

Principle 1: on "Secure Systems"

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SECURE SYSTEMS DO NOT EXIST

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

Capire chi è il nemico è importante per comprendere il livello di sicurezza adeguato.

What are we protecting? From who?

Principle 1: on "Secure Systems"

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SECURE SYSTEMS DO NOT EXIST

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

Quali sono le motivazioni di un possibile attacco?

What are we protecting? From who? Why?

Principle 1: on "Secure Systems"

Pensare in termine di valore perchè è il modo corretto per valutare il livello di sicurezza del sistema

SECU…XIST

- The **security le**… the **amount of time** necessar…**ey** (or resources) an…

- **Most times** the adversary is **rational**
- **Most times** the adversary has an **economic goal** (that is… **$$$**)
- **This is not always true** when the adversary is a **government**

What are we protecting? From who? Why?

Principle 1: on "Secure Systems"

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- This is an "informal way" for defining a THREAT MODEL
- Threat modeling is a structured process used in cybersecurity to identify and analyze potential threats and vulnerabilities in a system or application

O NOT EXIST

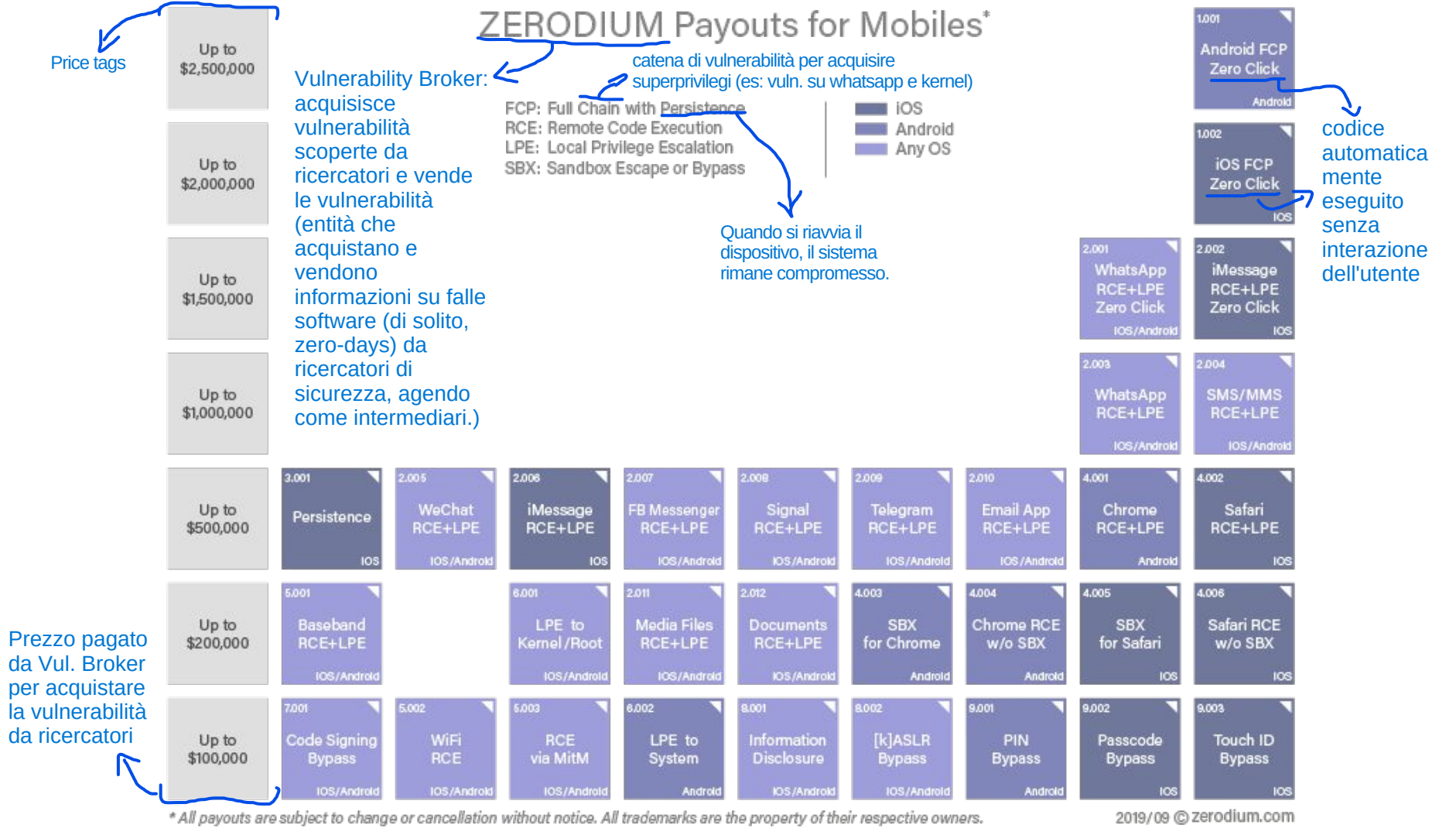
Definendo il Threat Model, sono in grado di definire il livello di sicurezza di cui il sistema ha bisogno

determined by the **amount of**

e **amount of money** (or

uccess

What are we protecting? From who? Why?


ZERODIUM Payouts for Mobiles*
Price tags
Vulnerability Broker: acquisisce vulnerabilità scoperte da ricercatori e vende le vulnerabilità (entità che acquistano e vendono informazioni su falle software (di solito, zero-days) da ricercatori di sicurezza, agendo come intermediari.)
catena di vulnerabilità per acquisire superprivilegi (es: vuln. su whatsapp e kernel)
FCP: Full Chain with Persistence
RCE: Remote Code Execution
LPE: Local Privilege Escalation
SBX: Sandbox Escape or Bypass
iOS
Android
Any OS
Quando si riavvia il dispositivo, il sistema rimane compromesso.
codice automaticamente eseguito senza interazione dell'utente
Prezzo pagato da Vul. Broker per acquistare la vulnerabilità da ricercatori
Up to $2,500,000
Up to $2,000,000
Up to $1,500,000
Up to $1,000,000
Up to $500,000
Up to $200,000
Up to $100,000
1.001 Android FCP Zero Click Android
1.002 iOS FCP Zero Click iOS
2.001 WhatsApp RCE+LPE Zero Click iOS/Android
2.002 iMessage RCE+LPE Zero Click iOS
2.003 WhatsApp RCE+LPE iOS/Android
2.004 SMS/MMS RCE+LPE iOS/Android
3.001 Persistence iOS
2.005 WeChat RCE+LPE iOS/Android
2.006 iMessage RCE+LPE iOS
2.007 FB Messenger RCE+LPE iOS/Android
2.008 Signal RCE+LPE iOS/Android
2.009 Telegram RCE+LPE iOS/Android
2.010 Email App RCE+LPE iOS/Android
4.001 Chrome RCE+LPE Android
4.002 Safari RCE+LPE iOS
5.001 Baseband RCE+LPE iOS/Android
6.001 LPE to Kernel/Root iOS/Android
2.011 Media Files RCE+LPE iOS/Android
2.012 Documents RCE+LPE iOS/Android
4.003 SBX for Chrome Android
4.004 Chrome RCE w/o SBX Android
4.005 SBX for Safari iOS
4.006 Safari RCE w/o SBX iOS
7.001 Code Signing Bypass iOS/Android
5.002 WiFi RCE iOS/Android
5.003 RCE via MitM iOS/Android
6.002 LPE to System Android
8.001 Information Disclosure iOS/Android
8.002 [k]ASLR Bypass iOS/Android
9.001 PIN Bypass Android
9.002 Passcode Bypass iOS
9.003 Touch ID Bypass iOS
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FCP: Full Chain with Persistence
RCE: Remote Code Execution
LPE: Local Privilege Escalation
SBX: Sandbox Escape or Bypass

iOS
Android
Any OS

While RCE allows attackers to execute commands from a remote location to a device, LPE allows a low-privileged user to gain administrative or root-level access.

A full-chain vulnerability on Mobile is a sophisticated cyberattack that links together multiple individual software flaws, often zero-day vulnerabilities, to achieve complete, unauthorized control over a device. While a single vulnerability might only cause an app to crash, a full chain links, for example, a browser exploit, a sandbox escape, and a kernel privilege escalation to move from an external attack to full "root" or kernel-level control.

Is not a full-chain vulnerability

WhatsApp RCE example:

CVE-2022-36934

Exploit Whatsapp to be able to run unauth code on the device and then doing something on the device itself (maybe there is something after that maybe another privilage escalation)

Up to $2,500,000								1.001 Android FCP Zero Click Android	
Up to $2,000,000									1.002 iOS FCP Zero Click iOS
Up to $1,500,000							2.001 WhatsApp RCE+LPE Zero Click iOS/Android		2.002 iMessage RCE+LPE Zero Click iOS
Up to $1,000,000							2.003 WhatsApp RCE+LPE iOS/Android		2.004 SMS/MMS RCE+LPE iOS/Android
Up to $500,000	3.001 Persistence iOS	WeChat RCE+LPE iOS/Android	iMessage RCE+LPE iOS	FB Messenger RCE+LPE iOS/Android	Signal RCE+LPE iOS/Android	Telegram RCE+LPE iOS/Android	2.010 Email App RCE+LPE iOS/Android	4.001 Chrome RCE+LPE Android	4.002 Safari RCE+LPE iOS
Up to $200,000	5.001 Baseband RCE+LPE iOS/Android	The WiFi HW use a Driver running in the Kernel Level	6.001 LPE to Kernel/Root iOS/Android	2.011 Media Files RCE+LPE iOS/Android	2.012 Documents RCE+LPE iOS/Android	4.003 SBX for Chrome Android	4.004 Chrome RCE w/o SBX Android	4.005 SBX for Safari iOS	4.006 Safari RCE w/o SBX iOS
Up to $100,000	7.001 Code Signing Bypass iOS/Android	5.002 WiFi RCE iOS/Android	5.003 RCE via MitM iOS/Android	6.002 LPE to System Android	8.001 Information Disclosure iOS/Android	8.002 [k]ASLR Bypass iOS/Android	9.001 PIN Bypass Android	9.002 Passcode Bypass iOS	9.003 Touch ID Bypass iOS

Automatic Privilage Excalation

Idea:
Exploit a Vulnerability in the Application and then exploit a Kernel Vulnerability to take control of Root Privilages

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Zerodium @Zerodium · Jun 1

We're looking for #0day exploits affecting Pidgin on Windows and Linux.

Bounty: $100,000

Read more: zerodium.com/temporary.html

9 129 213

Gary Kramlich
@rw_grim

Replying to @Zerodium

This really shows the sad state of funding in open source software.. I managed to find 25k USD last year to work on @impidgin full time, but if you can find a security exploit in my and other's non paid work you can make 4 times as much as I did... 💪

9:59 PM · Jun 1, 2021 · Twitter Web App

Most of the time, the attacker is rational

The dev can introduce the vuln. and then selling them

A lot of software products rely on open-source libraries

Idea Cloud Computing:
Exploit App Vulnerability, Exploit Virtual OS Kernel Vulnerability and Exploit Virtualization Software Vulnerability (to escape the VM) to arrive on the HW or OS of the Machine

Vuln. Broker are against the Fix of the Vulnerability (the Manifacture wants to fix it)

1) Sell the Vuln to Manifacture (Participate to Bug Bounty Program: get way less money than from Vuln. Brokers)
2) Sell the Vuln to Vuln. Broker

Principle 1: on "Complexity of Systems"

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MORE COMPLEXITY → LESS SECURITY

- (***Empirical rule***) **Increasing the complexity** of a systems leads to an **insecure system**
- **Linear increase** of complexity means **super-linear increase** of insecurity

Principle 1: on "Complexity of Systems"

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MORE COMPLEXITY → LESS SECURITY

- Do you want to build a "***reasonably secure***" system?

Reduce the Complexity of the System

KISS RULE:

- ***Keep It Simple and Stupid***
- *Keep It Simple, Stupid*
- *Keep It Short and Stupid*

If you don't understand the whole system, it is very difficult to have that system secure

Principle 1: on "Complexity of Systems"

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MORE COMPLEXITY → LESS SECURITY

- Does this mean that the systems to be secured can **not** be **complex** or **complicated**?

Principle 1: on "Complexity of Systems"

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MORE COMPLEXITY → LESS SECURITY

- Does this mean that the systems to be secured can **not** be **complex** or **complicated**?

NO!

Principle 1: on "Complexity of Systems"

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MORE COMPLEXITY → LESS SECURITY

- The **protection system** must be "***as simple as possible***". If possible... "***very simple***"
  - It is the most important part of the whole system (example: Authentication of the device to use it)
- **Unnecessary complexity** can lead to {**design**, **implementation**, **usage**} errors
- Each **error** is a potential "**vulnerability**"

Principle 2: on “Entities composing a system”

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Zones of the system where we can have a Vulnerability

(MOST) SYSTEMS ARE COMPOSED OF 3 TYPES OF ENTITIES

- Software
- Hardware (physical security is garanteed if bad guy isn't able to put hands on the HW)
- Humanware People inside the system

Principle 2: on "Entities composing a system"

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SOFTWARE

- It is a critical part of the system with a **large exposed attack surface**
- remember that... **often other parts of the system are "easier to attack"**

Princ…ng a system”

- **Many**, **very complex programs**
- **Each program made of thousands or millions of lines of source code (*i.e. computer instructions*)**

…RE

- It is a critical part of the system with a **large exposed attack surface**
- remember that… **often other parts of the system are “easier to attack”**

Principle 2: on "Entities composing a system"

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SOFTWARE

Air-Gapped Computer: This refers to a computer system that is physically isolated from all insecure networks, including the internet. The name implies there is a "gap of air" between the machine and any network, offering maximum security against remote, network-based, or malware attacks.

- It is a critical part of the system with a **large exposed attack s**
- r… the system are "**easier to attack**"

That can be accessed remotely (i.e. from the Internet, from a wireless network or from a user interface)

(example: cars)

Principle 2: on "Entities composing a system"

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SOFTWARE

doing a well attack suface analysis isn't so easy

- It is a critical part of the system with a **large exposed attack surface**

tech or not (example: a paper document containing information about a company is considered a significant part of an organization's physical attack surface)

- rememl ***Everything* that can be used by an attacker to violate the system** tem are "**easier to attack**"

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SOFTWARE

- It is a critical part of the system with a **large exposed attack surface**
- remember that... **often other parts of the system are "easier to attack"**

Principle 2: on "Entities composing a system"

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Attacking the software often requires some technical skills… attacking the human in the system requires skills that are not about computing (psychological skills. Example of attack: phishing)

OFTWARE

stem with a **large exposed attack**

- remember that… **often other parts of the system are "easier to attack"**

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HARDWARE

When firmware cannot be updated to remediate vulnerabilities, organizations must shift from patching to mitigation by removing the hardware or reducing its attack surface to manage risks. Firmware vulnerabilities are high-impact because they operate below the operating system level, often allowing for persistent, stealthy access.

- All the modern hardware is composed of **hardware AND software** (that is called **firmware**)
- The hardware can be **tampered** by an attacker
- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

Principle 2: on "Entities composing a system"

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HARDWARE

- All the modern hardware is composed of **hardware AND software** (that is called **firmware**)
- acker
- nd devices (**e.g. to deny** tee

This means that an attacker can modify the hardware or act on the software (e.g. flashing a modified firmware)

if the firmware is updatable and not well secured, the attacker can install a firmware with a backdoor

Principle 2: on "Entities composing a system"

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Definition: "*interfere with (something) in order to cause damage or make unauthorized alterations*"

ARE

…posed of **hardware AND software**

physical manipulation of a device's components, circuitry, or logic with malicious intent. Tampering requires the attacker to have physical access with the hardware.

- The hardware can be **tampered** by an attacker

How to know if an HW have been tampered? To garantee low-tampering HW, you need to garantee the physical security of the device (because if an HW has been tampered, it is very difficult to know it)

- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

Principle 2: on "Entities composing a system"

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Modern CPUs incorporate hardware-based anti-tamper mechanisms, such as cryptographic firmware signing, to avoid unauthorized modifications.

HARDWARE

The Physical security of an HW must be guaranteed across the full hardware lifecycle, from initial purchase to final disposal (because if control were lost even for just a few seconds, it could be tampered). We do this because it is very difficult to detect if the HW it was tempered

The "evil maid attack" is an example of unauthorized alteration of personal devices

- composed of **hardware AND software**
- The hardwar e **tampered** by an attacker
- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

Principle 2: on "Entities composing a system"

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Possible to put a backdoor in a binary firmware without knowing the source code (it easily to place it in the midlle of the firmware)

HARDWARE

It is an attack that exploits temporary physical access to compromise a device so that the attacker can control it remotely (by inserting a backdoor into the device's firmware).

The "evil maid attack" is an example of unauthorized alteration of personal devices

Evil Maid Attack Demo

A "nice demo": the time required to flash the tampered firmware is very limited

- The hardwar[illegible]e **tampered** by [illegible]acker
- Even the **physical security** of syst[illegible]ms and devices (**e.g. to deny unauthorized access**) is hard to guarantee

Principle 2: on "Entities composing a system"

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HUMANWARE

- The "**human in the loop**" is often the **weakest link in the system** (e.g. **social engineering**)
- For **many reasons**: lack of knowledge, overload, fatigue, bad design, goodwill, lack of attention… stupidity (*in both **users** and **system administrators***)

Principle 2: on "Entities

Definition: "the use of deception to manipulate individuals into divulging confidential or personal information that may be used for fraudulent purposes" (*e.g. phishing emails*)

inganno

to push him/her

(that goes against cybersecurity policies)

Social engineering is widely recognized as the most common and effective method for cyberattacks in the history of cybersecurity. According to industry statistics, social engineering techniques are involved in 98% of cyberattacks, making the "human element" the weakest link in security.

- The "**human in the loop**" is often the **weakest link in the system** (e.g. **social engineering**)

- For **many reasons**: lack of knowledge, overload, fatigue, bad design, goodwill, lack of attention… stupidity (*in both **users** and **system administrators***)

sovraccarico (può essere indotto dall'attaccante)

of tools

In such cases, people tend to cut corners by failing to follow cybersecurity policies and this makes the system vulnerable

Principle 2: on "Entities composing a system"

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HUMANWARE

Correct way to mitigate this problem is throgh procedures (properly designed) to identify if i'm talking with a collegue or impostor

buona volontà
Is "goodwill" a problem?
(example: faking to be a collegue and ask for confidential info)

Yes, when abused by attackers

- The "**human in the l** **he system** (e.g. **social enginee**
- For **many reasons**: lac knowledge, overload, fatigue, bad design, goodwill, lack of attention… stupidity (*in both **users** and **system administrators***)

We are stupid when we don't understand something

Principle 2: on “Entities composing a system”

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Operational Security (OPSEC) is the process of protecting sensitive information by looking at your own operations through the eyes of an attacker.

SECURITY AS A PROCESS

Cybersecurity isn't a product

- Securing a system is a [**never ending**] **process** and **not a product**

(Cybersecurity is a day-by-day investment)

- **Reasonably secure system** requires **an endless work** on the system and **education**

secure

1. The system that today is “secure”, tomorrow will not be. **Everyday news faults are discovered**
2. The **lack of updates** leads to **fragile** and **insecure** systems (system can be compromised)

Principle 2: on "Entities composing a system"

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SysAdmin focus on Updates (when to do it, how, etc..)

EVEN UPDATING IS NOT EASY...

- Are the updates **available** for all your devices? (e.g. **smartphone**)
- In other words, **is your devices supported**? **Until when**? → if the device is no longer supported, the device is vulnerable if they discover a vulnerability and you cannot update it

Principle 2: on "Entities composing a system"

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EVEN UPDATING IS NOT EASY...

- Are the updates **available** for all your devices? (e.g. **smartphone**)
- In other words, **is your devices supported**? **Until when**?
- If the updates are available...

Based on Company Context → Update Policy

The tech guy need to define the Update Policy

  - ***what*** *update to apply?* (***All of them***... *are you sure of that?*) Choose which part of the update to do
  - ***when*** *do you apply updates?* (*During the* ***night****? In "****production****"?*) if the vuln. is public known, update ASAP, considering the time (example: customer peeks, etc..)
  - ***why****?* (*Is that for* ***security*** *or to add* ***new functionalities****?*)
  - ***how****?* (*What is the best way to apply the update?* Automatic or ***Unattended****?*)

Principle 2: on "Entities composing a system"

EVEN UPDATING IS NOT EASY…

- The **updating process** is always a dangerous operation
  - for example: ***fixing bugs*** *can introduce* ***new bugs***, ***instability***, ***side-effects***… due to complexity of the system we are dealing with
- What we need is a **costs/benefits evaluation**
  - that is **costly** (*e.g. time consuming*)
  - it is often based on **partial information** (*i.e. often there is a lack of detailed information on the* ***updates content***) to take decision on a specific update

Principle 3: on “Security = Knowledge”

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KNOWLEDGE OF THE SYSTEM

Trust is even more important than Knowledge

- Real security is based on **deep knowledge** (of the system to be protected)
- What are the **effects** of this principle **on software**?
  - **Open Systems** vs. **Closed Systems**
  - **Open Source** vs. **Closed Source**

Principle 3: on "Security = Knowledge"

KNOWLEDGE OF THE SYSTEM

- Real security is based on **dee** be protected)
- What are the **effects** of this pri ple **on software**?
  - **Open Systems** vs. **Closed Systems**
  - **Open Source** vs. **Closed Source**

The specifications of the system are

OPEN / CLOSED

owledge”

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Prof. Opinion

I really prefer OPEN specifications (e.g. for *interoperability* with other system) but this does not guarantees that the system is secure (e.g. due to design or implementation faults)

E OF THE SYSTEM

- Real securit ed on dee be protected)
- What are t e **effects** of this pri ple **on software**?
  - **Open Systems** vs. **Closed Systems**
  - **Open Source** vs. **Closed Source**

The specifications of the system are

OPEN / CLOSED

If they have specs open/closed, doesn't guarantee security by definition

Principle 3: on "Security = Knowledge"

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KNOWLEDGE OF THE SYSTEM

- Real security is based on **deep knowledge** (of the system to be protected)
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Principle 3: on "Security = Knowledge"

KNOWLEDGE OF THE SYSTEM

- Real security is based on d… be protected)
- What are the **effects** of this prin… **software**?
  - **Open Systems** vs. **Cl… Systems**
  - **Open Source** vs. **Closed Source**

The source code (i.e. computer instructions) used by the programmers to implement the system are

OPEN / CLOSED doesn't guarantee security by definition

Prof. Opinion

I really prefer OPEN Source (*e.g. it can be inspected, verified, validated etc.*) but this does not guarantees that the system is secure

Knowledge"

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GE OF THE SYSTEM

- Real secu sed on d be protectec
- What are effects of this prin software?
  - Open ystems vs. Cl Systems
  - Open Source vs. Closed Source

The source code (i.e. computer instructions) used by the programmers to implement the system are

OPEN / CLOSED

I really prefer **OPEN** Source (*e.g. it can be inspected, verified, validated etc.*) but this does **not guarantees that the system is secure**

Even if it is **theoretically possible** (for me) **to check all the code in a system** (*e.g. an open source Operating System like GNU/Linux*), **this is not practically possible** (*since it is made of millions of lines of source code*)

Even if i have the source code of the program, it is difficult to know if the binary version of it contains a backdoor or malicious code that is different from the source code that is public available

- Real secu sed on m to be protecte
- What are effec of this principle on software?
  - **Open ystems** vs. **Closed Systems**
  - **Open Source** vs. **Closed Source**

Principle 3: on "Security = Knowledge"

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Management of Trust is fundamental in Cybersecurity and based on a lot of different stuff (example: choosing the browser based on history of vuln. fixing)

KNOWLEDGE OF THE SYSTEM

- Real security is based o… knowledge (of the system to be protected)
- What are the **effec…**
  - **Open Systems**
  - **Open Source** v…

In most cases it is a matter of

Security is based on Knowledge about the system we want to secure

baseline of CySec (define the allies and enemies)

TRUST

more important than knowledge (because it is very difficult to know/verify/check every single detail of the system)

Do you prefer to trust a community (i.e. Open Source) or a company (i.e. Close Source)

The future in terms of guaranteeing the supply-chain is a game not based on verifying everything, it will be based a lot on certifications (made in-house or by third-party certifiers) to demonstrate to customer that you are implementing cybersecurity

Nobody, now, create SW that is totally developer e/o designed in house (we use a lot of library, sw made by others, etc..): possible problem of the supply chain (example: (supply-chain attack) possible backdoor in our program that came from library developed by others)

Auto-Updates is a pro and cons:
- Pro: install automatically the fix for a detected vuln.
- Cons: install updates automatic with backdoors or other vuln.

Product of a Manifacture is based on Software supply chain (built on libraries, services, etc..): if something bad happens the responsability is on the Manifacture (this is why they need to certificate them to build trust on Manifacture's product)

Principle 3: on "Security = Knowledge"

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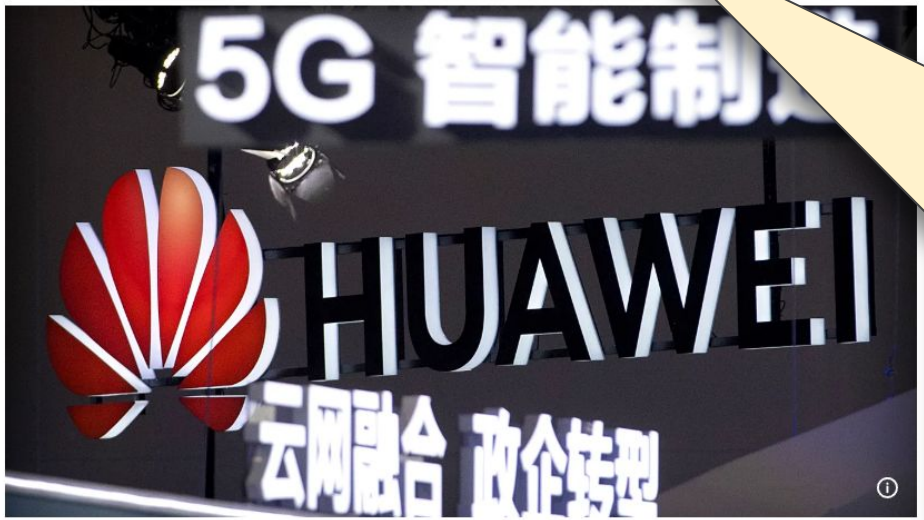
Eleven EU countries took 5G security measures to ban Huawei, ZTE



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By Cynthia Kroet
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because they cannot go in deep on the software of these devices to check about vulnerabilities like backdoors (they restricted because they don't have trust on some companies: possible that China gather info through it (not verified))

«Eleven countries [...] have used legal powers to impose **restrictions** on telecom suppliers that are considered high-risk, such as Huawei and ZTE, for 5G network infrastructure, a European Commission spokesperson told Euronews»

Principle 3: on "Security = Knowledge"

KNOWLEDGE OF THE SYSTEM

- **Users' education** is a form of **knowledge**
- **Investing in education** is good but **not that simple**:
  - education is **costly**
  - **perception of security** [lack of] importance
  - [wrong] **habits** (*i.e.* "*we have always done it this way*")
  - **ideological resistance** (*i.e.* "*information anarchism*")
  - **cost of security** procedures (*vs.* "*productive investments*")

Addendum: "Security of a Cryptosystem"

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KERCKHOFFS'S PRINCIPLE

- "A **cryptosystem** should be secure **even if everything about the system**, **except the key**, **is public knowledge**" (1883)

Addendum: "Security of a Cryptosystem"

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KERCKHOFFS'S PRINCIPLE

- "A **cryptosystem** should be secure **even if everything about the system, except the key, is public knowledge**" (1883)

It is a mechanism (i.e. hardware or software) used for encrypting information with the aim to preserve confidentiality

Addendum: "Security of a Cryptosystem"

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SHANNON'S REFORMULATION

- "The **enemy knows the system**"

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SECURITY THROUGH OBSCURITY

- Means relying on the **design** or **implementation secrecy** as the **main method of providing security**

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SECURITY THROUGH OBSCURITY

- Means relying on the **design** or **implementation secrecy** as the **main method of providing security**
- The **security through obscurity** is **WRONG**

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The **security through obscurity** is **WRONG**

- Does this means that I must **publish all the details about my system internals?**

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The **security through obscurity** is **WRONG**

- Does this means that I must **publish all the details about my system internals?**
- **NO**, it would be very stupid!

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CyberSecurity

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Attributions and Notes

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- These slides are partially based on the work of Prof. **Renzo Davoli**
- **Ludovico Gardenghi** has worked on a previous version of these slides
- I wish to **thank all of them**, **any errors that remain are my sole responsibility**